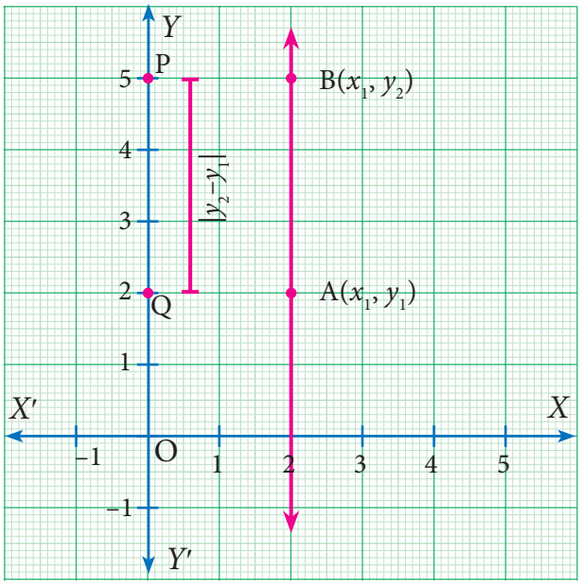



Fig. 5.17

$A(x_1, y_1)$ and $B(x_1, y_2)$, parallel to $y$ – axis.

Then the distance between these two points is

$$|y_2 - y_1|$$

[The difference between y coordinates]

### 5.3.3 Distance Between Two Points on a Plane.

Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ be two points in the Cartesian plane (or $xy$ – plane), at a distance '$d$' apart that is $d = PQ$.

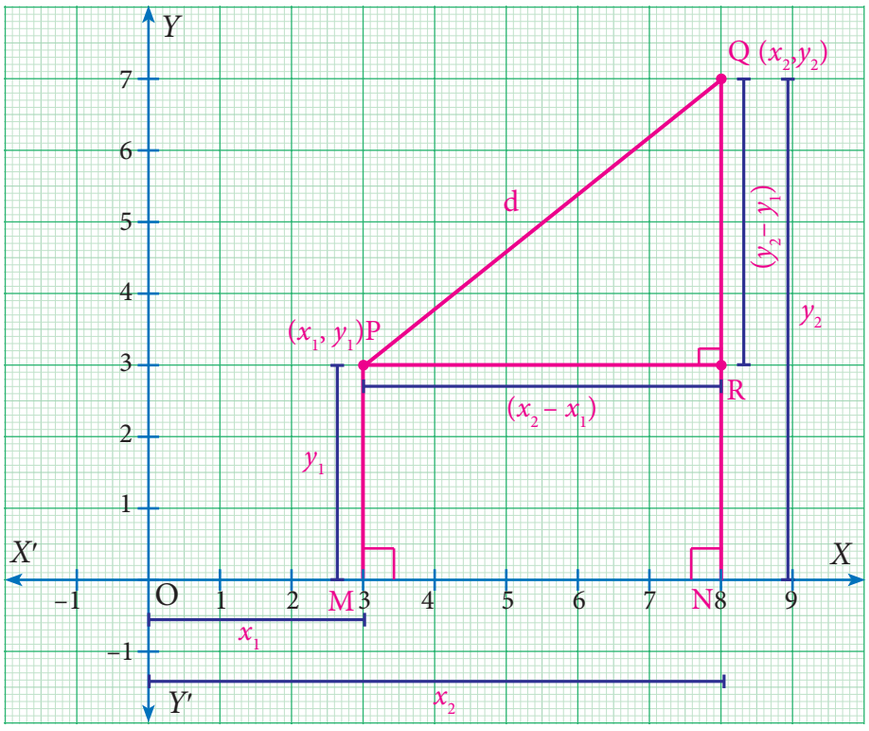



Fig. 5.18

**Step 1**

By the definition of coordinates,

$OM = x_1$ $\quad MP = y_1$

$ON = x_2$ $\quad NQ = y_2$

Now, $(PR \perp NQ)$

$PR = MN$ (Opposite sides of the rectangle $MNRP$)

**Thinking Corner**

A man goes 3 km towards north and then 4 km towards east. How far is he away from the initial position?



5-Coordinat_1.indd 196 02/12/2022 21:44:44